

Multimodal Road Accident Detection System

Problem Statement

India records over 50,000 hit and run cases every year. In most cases the accused vehicle escapes because there is no automated system to instantly capture evidence. Existing CCTV cameras only record footage passively — someone has to manually watch hours of video to find the accident moment. By that time the vehicle has already fled and evidence is lost. There is no open automated system that can analyze road footage, detect an accident, confirm it through audio, capture the evidence frame, and extract the vehicle number — all together without any human intervention.

Proposed Solution

A multimodal pipeline that accepts a road accident video as input and automatically processes all four modalities to generate a structured PDF evidence report:

- **Video:** Detect vehicle collision from CCTV footage using YOLOv8
 - **Audio:** Confirm accident through crash sound and screech detection using Librosa
 - **Image:** Capture accident frame as photograph evidence using OpenCV
 - **Text:** Extract vehicle number plate from pre-impact frames using EasyOCR
- Output:** A downloadable PDF evidence report containing accident timestamp, audio confirmation, vehicle photograph, extracted number plate, and severity level.